

Probability is Lack of Information: Part IV The Second Derivative of Binary Entropy as the Curvature of Determinism

Murat Hüsnü Sazlı, Ph.D.

Professor and Head of Telecommunications Chair

Department of Electrical and Electronics Engineering, Faculty of Engineering

Ankara University, Gölbaşı, Ankara, Türkiye

Correspondence: sazli@eng.ankara.edu.tr

June 21, 2026

Abstract

Following the axiomatic foundation established in Part I, the formal proof via the Binary Entropy Function in Part II, and the introduction of the first derivative as the Gradient of Determinism in Part III, this fourth installment examines the second derivative of the binary entropy function. We demonstrate that $H''(p) = -\frac{1}{p(1-p)\ln 2}$ quantifies the curvature of determinism, revealing the rate at which the gradient of determinism itself changes.

This second derivative is always negative, confirming the inherent concavity of the entropy function, with its most negative value at $p = 0.5$ – the point of maximum epistemic uncertainty. This region represents the highest “acceleration” toward deterministic resolution. We interpret this curvature as a dynamic measure of how rapidly informational deficit collapses as observational resolution increases within the Hendeca-Tier Cosmic Signal Architecture (HCSA).

By extending the analysis from gradient to curvature, we provide a more complete mathematical framework for understanding the transition from epistemic probability to ontological determinism. This installment strengthens the central thesis of the series: probability is not a fundamental property of nature, but a measurable indicator of missing information whose rate of resolution can be rigorously quantified.

The philosophical motto of the series continues to guide our inquiry: **Dubito ergo cogito ergo sum $\rightarrow$ Dubito ergo determinismus est. I doubt, therefore I think, therefore I am — I doubt, therefore determinism exists.**

Keywords: Probability is Lack of Information, Binary Entropy Function ($H(p)$), Second Derivative of Binary Entropy, Curvature of Determinism, Derivative of Binary Entropy, Binary Entropy Derivative, Entropy Gradient, Gradient of Determinism, Universal Determinism, Information-Theoretic Benchmarking, Formal Proof, Horse-shoe Geometry, Informational Deficit, Cevher-i Ferd Limit, Hendeca-Tier Cosmic Signal Architecture (HCSA), 121-Unit Mizanic Scale, Stochastic-to-Deterministic Transition, Signal Resolution.

1 Introduction

The series “Probability is Lack of Information” continues its systematic deconstruction of the probabilistic paradigm. In Part I, the foundational axiom was established: probability is not an ontological feature of reality but rather a measure of the observer’s informational deficit [1]. In Part II, this axiom received its formal validation through the geometric and differential properties of the Binary Entropy Function $H(p)$, demonstrating that the celebrated “horse-shoe” curve is in fact a precise map of the distance between limited human observation and the underlying deterministic architecture of the universe [2]. In Part III, the first derivative of the binary entropy function was introduced as the **Gradient of Determinism**, providing a dynamic measure of how rapidly informational uncertainty collapses into certainty [3].

Building upon these foundations, the present Part IV examines the **second derivative** of the binary entropy function. While the first derivative quantifies the *rate* of transition from epistemic probability to ontological determinism, the second derivative reveals the *curvature* of this transition – the rate at which the gradient itself changes. This introduces a higher-order dynamical perspective into the analysis.

Mathematically, the second derivative is given by

$$H''(p) = -\frac{1}{p(1-p)\ln 2} \quad (1)$$

which is always negative, confirming the inherent concavity of the entropy function. Its most negative value occurs at $p = 0.5$, the point of maximum epistemic uncertainty, indicating the region where the transition toward determinism exhibits the highest acceleration.

At the heart of this work lies a fundamental reinterpretation: the curvature of binary entropy offers a rigorous mathematical instrument to understand not only *how much* uncertainty exists, but *how rapidly* it resolves as observational resolution improves. This dynamic view provides deeper insight into the nature of determinism within the Hendeca-Tier Cosmic Signal Architecture (HCSA), where the observer’s limited 61-unit window interacts with the complete 121-unit mizanic structure of reality.

This installment also reinforces the philosophical motto that captures the epistemic spirit of the entire series. Building upon the profound insight of René Descartes, who established the indubitable foundation of modern philosophy with his famous “*Dubito ergo cogito ergo sum*” (I doubt, therefore I think, therefore I am), we extend this chain of certainty into the realm of cosmic reality:

Dubito ergo cogito ergo sum $\rightarrow$ *Dubito ergo determinismus est.*
I doubt, therefore I think, therefore I am — I doubt, therefore determinism exists.

The remainder of this paper is organized as follows. Section 2 presents the mathematical derivation and properties of the second derivative. Section 3 explores its interpretation as the curvature of determinism. Section 4 discusses broader implications for quantum mechanics and information theory. Finally, Section 5 concludes the paper and provides an outlook for subsequent installments in the series.

2 The Second Derivative of Binary Entropy as the Curvature of Determinism

While the first derivative of the binary entropy function provides a dynamic measure of the transition from epistemic uncertainty to determinism, the second derivative reveals the **curvature** of this transition – how rapidly the gradient itself changes. This higher-order analysis introduces a new layer of dynamical insight into the nature of determinism.

2.1 Mathematical Derivation

The binary entropy function for a Bernoulli process is given by

$$H(p) = -p \log_2(p) - (1 - p) \log_2(1 - p), \quad 0 < p < 1 \quad (2)$$

The first derivative, as established in Part III, is

$$H'(p) = \log_2 \left(\frac{1 - p}{p} \right) \quad (3)$$

Differentiating once more with respect to p yields the second derivative:

$$H''(p) = -\frac{1}{p(1 - p) \ln 2} \quad (4)$$

This expression is always negative for $p \in (0, 1)$, confirming that $H(p)$ is a strictly concave function.

2.2 Key Properties of the Second Derivative

The second derivative $H''(p)$ exhibits several important characteristics:

1. **Always Negative:** $H''(p) < 0$ for all $p \in (0, 1)$, which mathematically proves the concavity of the binary entropy function.
2. **Maximum Curvature at $p = 0.5$:** The most negative value occurs at $p = 0.5$, where

$$H''(0.5) = -\frac{4}{\ln 2} \approx -5.77$$

This corresponds to the point of maximum epistemic uncertainty, where the rate of change of the determinism gradient is highest.

3. **Asymptotic Behavior:** As $p \rightarrow 0^+$ or $p \rightarrow 1^-$, $H''(p) \rightarrow -\infty$. Near the deterministic boundaries, the curvature becomes extremely sharp.
4. **Units:** The second derivative has units of **bits per (probability)²**, representing the acceleration of informational resolution.

Figure 1 illustrates the behavior of $H''(p)$ across $p \in (0, 1)$. The curve is symmetric around $p = 0.5$, reaches its minimum there, and diverges to $-\infty$ at the boundaries.

2.3 Interpretation within HCSA

Within the **Hendeca-Tier Cosmic Signal Architecture (HCSA)**, $H''(p)$ is reinterpreted as the **Curvature of Determinism**. It measures how rapidly the gradient of determinism changes as the observer's informational resolution improves.

- At $p \approx 0.5$, the high negative curvature indicates a region of **maximum epistemic instability**, where small changes in probability lead to rapid shifts toward determinism.
- Near the boundaries ($p \rightarrow 0$ or $p \rightarrow 1$), the extreme curvature reflects near-instantaneous collapse of any remaining uncertainty.

This curvature thus provides a precise mathematical tool to distinguish not only the *degree* of determinism but also the *speed* at which a system moves toward it.

In the next section, we explore the deeper implications of this curvature for understanding the dynamics of determinism.

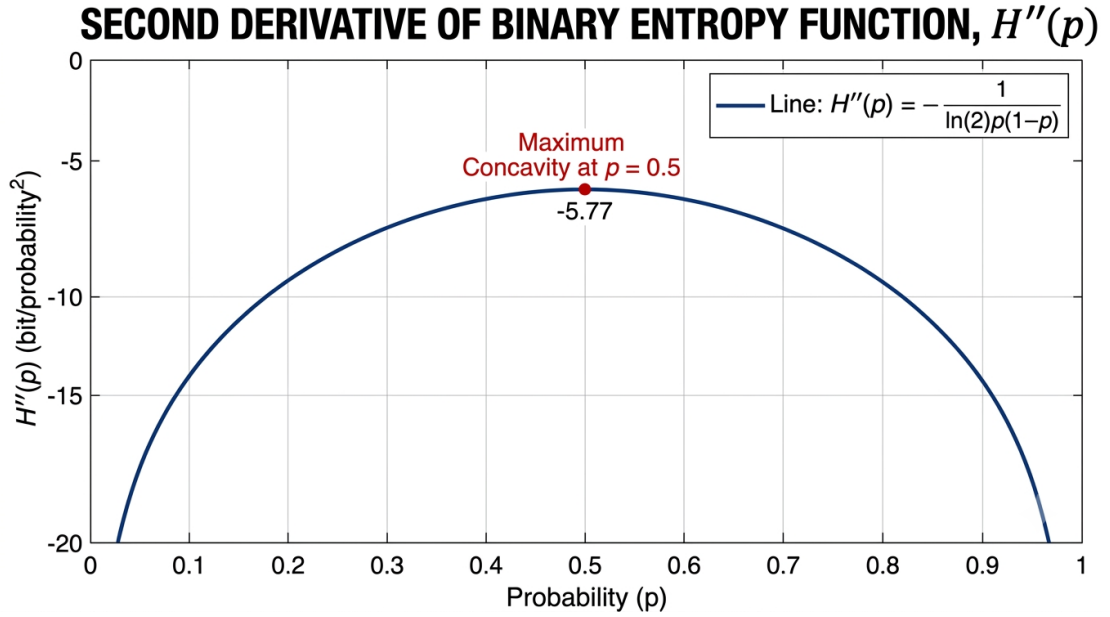


Figure 1 illustrates the behavior of $H''(p)$ across $p \in (0, 1)$. The visualization clearly shows the symmetric curvature, with the strongest negative value at the point of maximum uncertainty, providing an intuitive understanding of the acceleration of the transition toward determinism.

3 The Curvature of Determinism and Its Implications

Having established the second derivative $H''(p)$ as the Curvature of Determinism, we now explore its deeper implications for understanding the dynamics of the transition from epistemic uncertainty to effective determinism.

3.1 The Region of Maximum Acceleration

The most significant feature of the second derivative is its minimum value at $p = 0.5$:

$$H''(0.5) = -\frac{4}{\ln 2} \approx -5.77$$

This point corresponds to the peak of the entropy function (maximum ignorance). Here, the curvature is strongest, meaning that small changes in probability lead to the

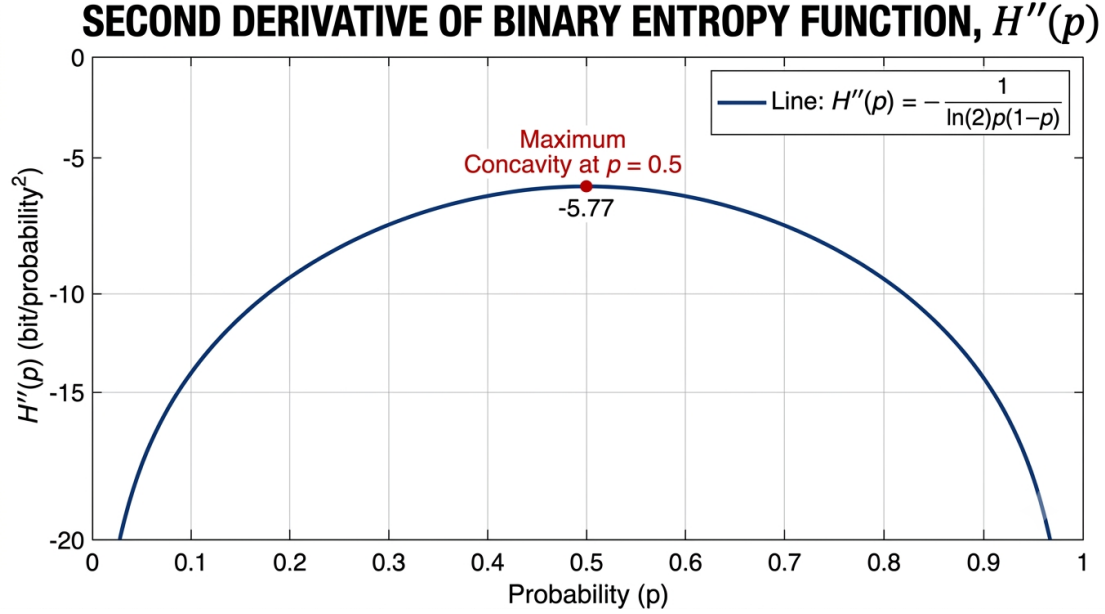


Figure 1: Second Derivative of the Binary Entropy Function $H''(p) = -\frac{1}{p(1-p)\ln 2}$. This curve represents the **Curvature of Determinism**. It reaches its minimum (most negative value) at $p = 0.5$, corresponding to the point of maximum epistemic uncertainty and highest acceleration toward determinism. The curve is symmetric and diverges to $-\infty$ as p approaches 0 or 1.

most rapid changes in the determinism gradient. In the context of the HCSA, this region can be interpreted as the **epistemic instability zone**, where the observer’s limited resolution encounters the highest “acceleration” toward deterministic resolution.

3.2 Dynamic Interpretation of Determinism

While the first derivative provides a static snapshot of determinism at a given probability, the second derivative adds a dynamic dimension:

- ****High Negative Curvature****: Indicates rapid collapse of uncertainty (strong acceleration toward determinism).
- ****Near-Zero Curvature****: Occurs away from $p = 0.5$, where the system is already in a relatively stable state.

This allows us to distinguish not only *how deterministic* a system is, but *how quickly* it is becoming deterministic.

3.3 Quantitative Thresholds for Curvature

We can extend the determinism thresholds from Part III by incorporating curvature:

Curvature Level	$ H''(p) $ Threshold	Interpretation
Moderate Curvature	$ H''(p) \geq 2$	Noticeable acceleration toward determinism
Strong Curvature	$ H''(p) \geq 4$	High acceleration (near critical transition)
Extreme Curvature	$ H''(p) \geq 5$	Maximum epistemic instability zone

Table 1: Proposed curvature-based thresholds for assessing the dynamics of determinism.

These thresholds complement the gradient thresholds and provide a more complete picture of the transition process.

In the following section, we discuss the broader implications of this curvature analysis for quantum mechanics and information theory.

4 Implications for Quantum Mechanics and Information Theory

The curvature analysis of the binary entropy function provides a powerful new mathematical lens through which to reinterpret foundational issues in quantum theory and information science.

4.1 The Curvature Perspective on Quantum Measurement

In quantum mechanics, the measurement problem has long been a central puzzle. The second derivative offers a fresh perspective: the apparent collapse of the wave function can be understood as the rapid change in the curvature of informational deficit. When a measurement is performed, the system moves away from the high-curvature region around $p = 0.5$ toward the deterministic boundaries, with the second derivative quantifying the acceleration of this process.

This view aligns with the idea that quantum probabilities reflect observer-limited information rather than fundamental indeterminism.

4.2 Born’s Rule Revisited

Born’s rule, which assigns probabilities as the square of the wave function amplitude ($|\psi|^2$), can be reinterpreted through the lens of entropy curvature. The transition from a superposition (high entropy, high negative curvature) to a definite state (low entropy) is characterized by a sharp change in $H''(p)$. This suggests that the “collapse” is not a mysterious physical process but the natural consequence of closing an informational gap, with the curvature describing the speed of this resolution.

4.3 Superdeterminism and Hierarchical Structure

Within the Hendeca-Tier Cosmic Signal Architecture (HCSA), the second derivative strengthens the superdeterministic interpretation. The high curvature at $p = 0.5$ explains why Bell-type experiments appear to violate statistical independence: the system is in a highly unstable epistemic state where small informational changes produce large dynamical effects. What appears as non-locality is simply the manifestation of shared informational structure across different observational tiers.

4.4 Toward a Dynamic Deterministic Framework

By incorporating both the gradient and the curvature, we move closer to a fully dynamic theory of determinism. Future work may explore whether the Schrödinger equation can be derived as an effective description emerging from the underlying mizanic dynamics when viewed through limited informational resolution. The entropy curvature thus serves

as both a diagnostic tool and a bridge between the epistemic limitations of the observer and the ontological determinism of the cosmic architecture.

5 Conclusion: Toward a Dynamic Theory of Determinism

This fourth installment of the “Probability is Lack of Information” series has extended the analysis from the first derivative (Gradient of Determinism) to the second derivative (Curvature of Determinism). By demonstrating that $H''(p) = -\frac{1}{p(1-p)\ln 2}$ quantifies the rate at which the gradient of determinism itself changes, we have provided a more complete dynamical framework for understanding the transition from epistemic uncertainty to ontological determinism.

The second derivative confirms the inherent concavity of the binary entropy function and identifies the region around $p = 0.5$ as the zone of maximum epistemic instability, where the acceleration toward deterministic resolution is highest. This curvature analysis complements the gradient thresholds introduced in Part III and offers new quantitative tools for distinguishing not only the degree but also the speed of the transition to determinism.

When combined with the foundational axiom of Part I, the geometric insights of Part II, and the gradient analysis of Part III, these results significantly strengthen the central thesis of the series: probability is not a fundamental property of nature, but a measurable indicator of missing information whose rate and acceleration of resolution can be rigorously quantified.

Within the **Hendeca-Tier Cosmic Signal Architecture (HCSA)**, the curvature of binary entropy serves as a critical diagnostic tool that bridges the observer’s limited 61-unit window and the complete 121-unit mizanic structure of reality. The philosophical motto of the series continues to capture this epistemic journey:

Dubito ergo cogito ergo sum $\rightarrow$ Dubito ergo determinismus est.
I doubt, therefore I think, therefore I am — I doubt, therefore determinism exists.

As we continue this series, subsequent installments will explore further mathematical and empirical implications of this framework for quantum information, cosmological observations, and the development of a fully deterministic Vertical Deterministic Operating System (VDOS). The transition from a probabilistic to a deterministic worldview is no longer a philosophical aspiration but a mathematically grounded inevitability.

Probability is lack of information. As informational resolution increases, determinism inevitably emerges.

Acknowledgments

The author acknowledges the critical role of advanced AI-based computational collaboration in the formalization, mathematical derivation, curvature analysis, and technical synthesis of the second derivative of the binary entropy function presented in this manuscript. This collaborative methodology has been instrumental in refining the concepts and visualizations.

The foundational conceptual framework of the **Hendeca-Tier Cosmic Signal Architecture (HCSA)**, the **121/11=11** symmetry, the **Gradient of Determinism**, the **Curvature of Determinism**, and the specific axiomatic derivations of the entire series remain the exclusive intellectual property of the author.

References

- [1] Sazli, M. H. (2026). *Probability is Lack of Information: The Information-Theoretic Foundation of Universal Determinism*. SSRN (April 20, 2026). Available at SSRN: <https://ssrn.com/abstract=6609460> or <http://dx.doi.org/10.2139/ssrn.6609460>.
- [2] Sazli, M. H. (2026). *Probability is Lack of Information: Part II -The Binary Entropy Function as a Formal Proof of Universal Determinism*. SSRN (April 27, 2026). Available at SSRN: <https://ssrn.com/abstract=6658599> or <http://dx.doi.org/10.2139/ssrn.6658599>.
- [3] Sazli, M. H. (2026). *Probability is Lack of Information: Part III -The Derivative of Binary Entropy as the Gradient of Determinism*. SSRN ID: 6962403. (Uploaded: June 18, 2026; In processing). Available at SSRN: <https://ssrn.com/abstract=6962403> or <http://dx.doi.org/10.2139/ssrn.6962403>.
- [4] Sazli, M. H. (2026). *Beyond Matter: The Hendeca-Tier Cosmic Signal Architecture (HCSA) and the Golden Centrality of Human Scale: A Multidisciplinary Synthesis of Information Theory, M-Theory, and Cosmic OSI Equivalence*. SSRN (January 16, 2026). Available at SSRN: <https://ssrn.com/abstract=6085046> or <http://dx.doi.org/10.2139/ssrn.6085046>.
- [5] Sazli, M. H. (2026). *Beyond Matter II: Deterministic Resolution of Dark Energy, Dark Matter, and Cosmic Microwave Background Anomalies via Hendeca-Tier Informational Pressure (10^{66}) A Theoretical Validation of the Hendeca-Tier Cosmic Signal Architecture (HCSA) as a Vertical Deterministic Operating System for the Universe: Bridging Kepler to Penrose in the Information Age*. SSRN (February 10, 2026). Available at SSRN: <https://ssrn.com/abstract=6209718> or <http://dx.doi.org/10.2139/ssrn.6209718>.
- [6] Sazli, M. H. (2026). *Beyond Matter III: Empirical Resolution of the James Webb Space Telescope (JWST) "Impossible Early Galaxies" via Hendeca-Tier Mizanic Scaling — The 10^5 Informational Compression Factor*. SSRN (March 1, 2026). Available at SSRN: <https://ssrn.com/abstract=6322859> or <http://dx.doi.org/10.2139/ssrn.6322859>.
- [7] Sazli, M. H. (2026). *Beyond Matter IV: Resolution of the Hubble Tension via Hierarchical Scaling Constant (HSC) and the 10^6 Dimensional Shift - The 6-Unit Mizanic Gap: Reconciling Planck and SH0ES Observations through the Hendeca-Tier Cosmic Signal Architecture (HCSA)*. SSRN (March 9, 2026). Available at SSRN: <https://ssrn.com/abstract=6375318> or <http://dx.doi.org/10.2139/ssrn.6375318>.
- [8] Born, M. (1926). Zur Quantenmechanik der Stoßvorgänge. *Zeitschrift für Physik*, 37(12), 863–867.